

Experiments

- Test hypothesis
- Predict the relationship between two or more variables.
- Independent variable is manipulated by the researcher.
- Dependent variable influenced by the independent variable.
- Typical experimental designs have one or two independent variables.
- Validated statistically & replicable.

Experimental designs

- Different participants - single group of participants is allocated randomly to the experimental conditions.
- Same participants - all participants appear in both conditions.
- Matched participants - participants are matched in pairs, e.g., based on expertise, gender, etc.

Different, same, matched participant design

Design	Advantages	Disadvantages
Different	No order effects	Many subjects & individual differences a problem
Same	Few individuals, no individual differences	Counter-balancing needed because of ordering effects
Matched	Same as different participants but individual differences reduced	Cannot be sure of perfect matching on all differences